COMPUTER NETWORKS

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Problem Statement:

Broadcast traffic in networks can lead to congestion and broadcast storms.
This project aims to control excessive broadcast packets using an SDN controller.
The controller limits broadcast traffic based on a predefined threshold.

1. Project Setup

```
(ryu-env) khushi@khushi:~/pox$ python3 --version
Python 3.12.3
(ryu-env) khushi@khushi:~/pox$ pip --version
pip 24.0 from /home/khushi/ryu-env/lib/python3.12/site-packages/pip (python 3.12)
(ryu-env) khushi@khushi:~/pox$
```

2. Controller Running (Terminal 1)

```
(ryu-env) khushi@khushi:~/pox$ ./pox.py log.level --DEBUG broadcast_control
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:broadcast_control:🚀 Initializing Broadcast Control Module
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 3 2026 12:15:18)
DEBUG:core:Platform is Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
DEBUG:openflow.of_01:1 connection aborted
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 2
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 3
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: de:40:64:3e:8e:8e | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from de:40:64:3e:8e:8e
INFO:broadcast_control:[Broadcast] Src: de:40:64:3e:8e:8e | Count: 2
INFO:broadcast_control:✅ Allowing Broadcast from de:40:64:3e:8e:8e
INFO:broadcast_control:[Broadcast] Src: 9a:45:ca:69:f7:2b | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from 9a:45:ca:69:f7:2b
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 4
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 5
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 6
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 7
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 8
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 9
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 10
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 11
WARNING:broadcast_control:🚫 BLOCKED Broadcast from a6:fa:38:9d:cf:ac (Limit Exceeded)
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 12
WARNING:broadcast_control:🚫 BLOCKED Broadcast from a6:fa:38:9d:cf:ac (Limit Exceeded)
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 13
```

3. Mininet Running (Terminal 2)

```
khushi@khushi:~$ sudo mn -c
sudo mn --topo single,4 --controller=remote
[sudo] password for khushi:
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
***  Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([-_.[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
```

4. Pingall Success

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
```

5. Normal Ping (Unicast)

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.80 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.10 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=3.23 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=3.28 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=4.02 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=3.16 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=3.15 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=3.11 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=3.81 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=4.05 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=3.60 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=3.11 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=3.31 ms
^C
--- 10.0.0.2 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12416ms
rtt min/avg/max/mdev = 3.109/3.517/4.098/0.374 ms
```

6.h1 ping -b 10.0.0.255

```
mininet> h1 ping -b 10.0.0.255
PING 10.0.0.255 (10.0.0.255) 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable
From 10.0.0.1 icmp_seq=2 Destination Host Unreachable
From 10.0.0.1 icmp_seq=3 Destination Host Unreachable
From 10.0.0.1 icmp_seq=4 Destination Host Unreachable
From 10.0.0.1 icmp_seq=5 Destination Host Unreachable
From 10.0.0.1 icmp_seq=6 Destination Host Unreachable
From 10.0.0.1 icmp_seq=7 Destination Host Unreachable
From 10.0.0.1 icmp_seq=8 Destination Host Unreachable
From 10.0.0.1 icmp_seq=9 Destination Host Unreachable
From 10.0.0.1 icmp_seq=10 Destination Host Unreachable
From 10.0.0.1 icmp_seq=11 Destination Host Unreachable
From 10.0.0.1 icmp_seq=12 Destination Host Unreachable
From 10.0.0.1 icmp_seq=13 Destination Host Unreachable
From 10.0.0.1 icmp_seq=14 Destination Host Unreachable
From 10.0.0.1 icmp_seq=15 Destination Host Unreachable
From 10.0.0.1 icmp_seq=16 Destination Host Unreachable
From 10.0.0.1 icmp_seq=17 Destination Host Unreachable
From 10.0.0.1 icmp_seq=18 Destination Host Unreachable
From 10.0.0.1 icmp_seq=19 Destination Host Unreachable
From 10.0.0.1 icmp_seq=20 Destination Host Unreachable
From 10.0.0.1 icmp_seq=21 Destination Host Unreachable
From 10.0.0.1 icmp_seq=22 Destination Host Unreachable
From 10.0.0.1 icmp_seq=23 Destination Host Unreachable
From 10.0.0.1 icmp_seq=24 Destination Host Unreachable
From 10.0.0.1 icmp_seq=25 Destination Host Unreachable
From 10.0.0.1 icmp_seq=26 Destination Host Unreachable
From 10.0.0.1 icmp_seq=27 Destination Host Unreachable
^C
--- 10.0.0.255 ping statistics ---
30 packets transmitted, 0 received, +27 errors, 100% packet loss, time 29058ms
pipe 3
```

7. Controller Logs

```
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 2
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 3
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: de:40:64:3e:8e:8e | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from de:40:64:3e:8e:8e
INFO:broadcast_control:[Broadcast] Src: de:40:64:3e:8e:8e | Count: 2
INFO:broadcast_control:✅ Allowing Broadcast from de:40:64:3e:8e:8e
INFO:broadcast_control:[Broadcast] Src: 9a:45:ca:69:f7:2b | Count: 1
INFO:broadcast_control:✅ Allowing Broadcast from 9a:45:ca:69:f7:2b
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 4
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 5
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 6
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 7
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 8
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 9
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 10
INFO:broadcast_control:✅ Allowing Broadcast from a6:fa:38:9d:cf:ac
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 11
WARNING:broadcast_control:🚫 BLOCKED Broadcast from a6:fa:38:9d:cf:ac (Limit Exceeded)
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 12
WARNING:broadcast_control:🚫 BLOCKED Broadcast from a6:fa:38:9d:cf:ac (Limit Exceeded)
INFO:broadcast_control:[Broadcast] Src: a6:fa:38:9d:cf:ac | Count: 13
```

8. Clean Command

```
(ryu-env) khushi@khushi:~/pox$ sudo mn -c
[sudo] password for khushi:
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_corelt-nox_core ovs-openflowd ovs-controllerovs-testcontroller udpbwtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
***  Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --if-exists del-br s1
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([-_.[:alnum:]]+-eth[[:digit:]]+)'
( ip link del s1-eth1;ip link del s1-eth2;ip link del s1-eth3;ip link del s1-eth4 ) 2> /dev/null
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
(ryu-env) khushi@khushi:~/pox$
```

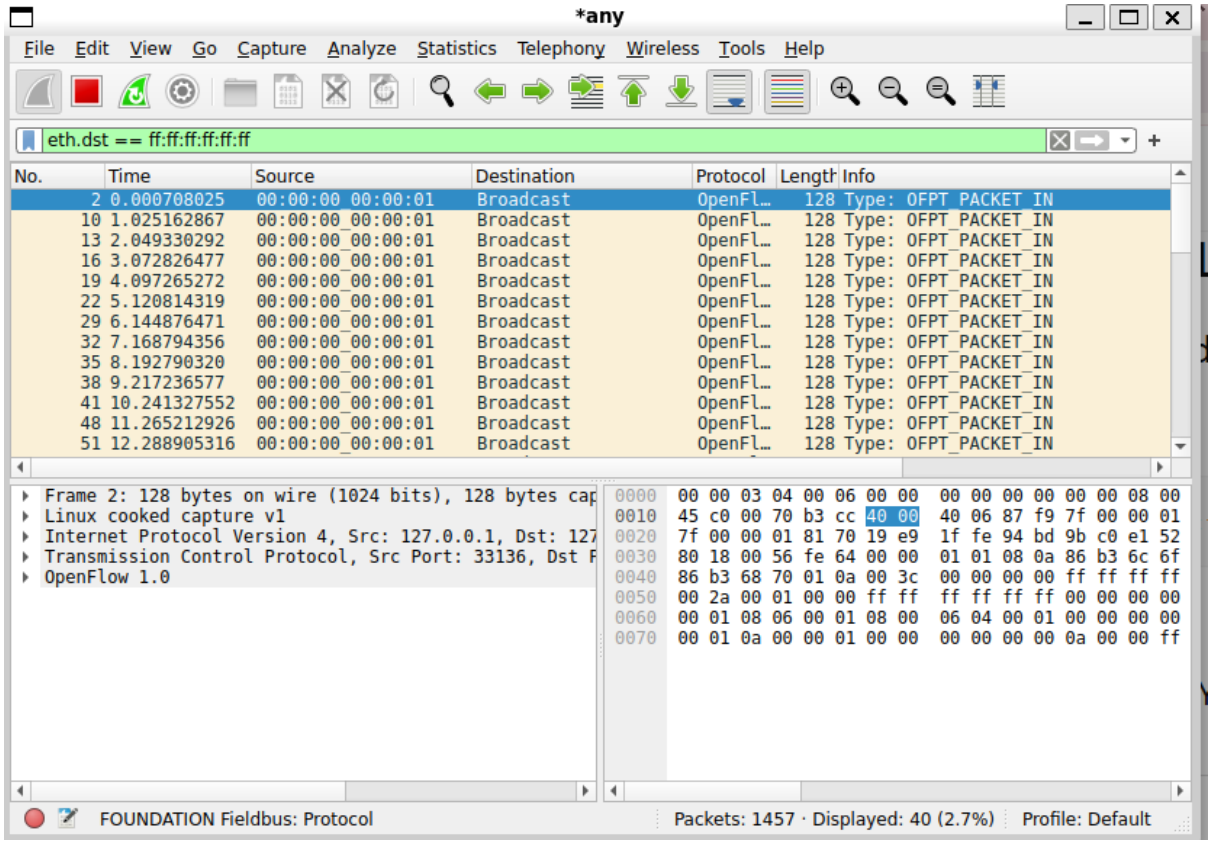
9.CODE:

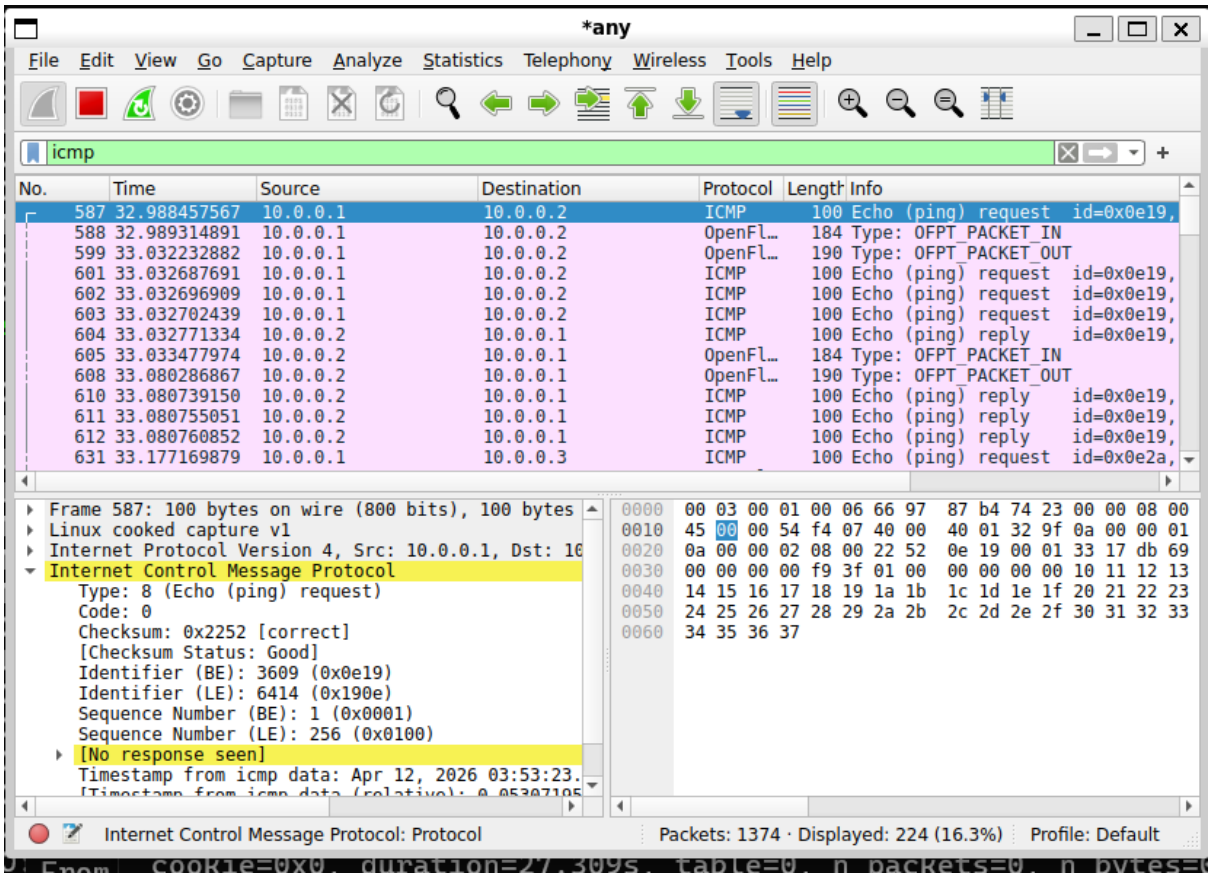
```
  GNU nano 7.2                                   /home/khushi/pox/pox/broadcast_control.py
from pox.core import core
import pox.openflow.libopenflow_01 as of

log = core.getLogger()

broadcast_count = {}
THRESHOLD = 10

def _handle_PacketIn(event):
    packet = event.parsed
    src = str(packet.src)
    dst = str(packet.dst)

    # Check if broadcast
    if dst == "ff:ff:ff:ff:ff:ff":
        broadcast_count[src] = broadcast_count.get(src, 0) + 1
        count = broadcast_count[src]

        log.info("[Broadcast] Src: %s | Count: %d", src, count)

        if count > THRESHOLD:
            log.warning("🚫 BLOCKED Broadcast from %s (Limit Exceeded)", src)
            return  # Drop packet

        else:
            log.info("✅ Allowing Broadcast from %s", src)

    # Forward packet
    msg = of.ofp_packet_out()
    msg.data = event.ofp
    msg.actions.append(of.ofp_action_output(port=of.OFPP_FLOOD))
    event.connection.send(msg)


def launch():
    log.info("🚀 Initializing Broadcast Control Module")
    core.openflow.addListenerByName("PacketIn", _handle_PacketIn)
```

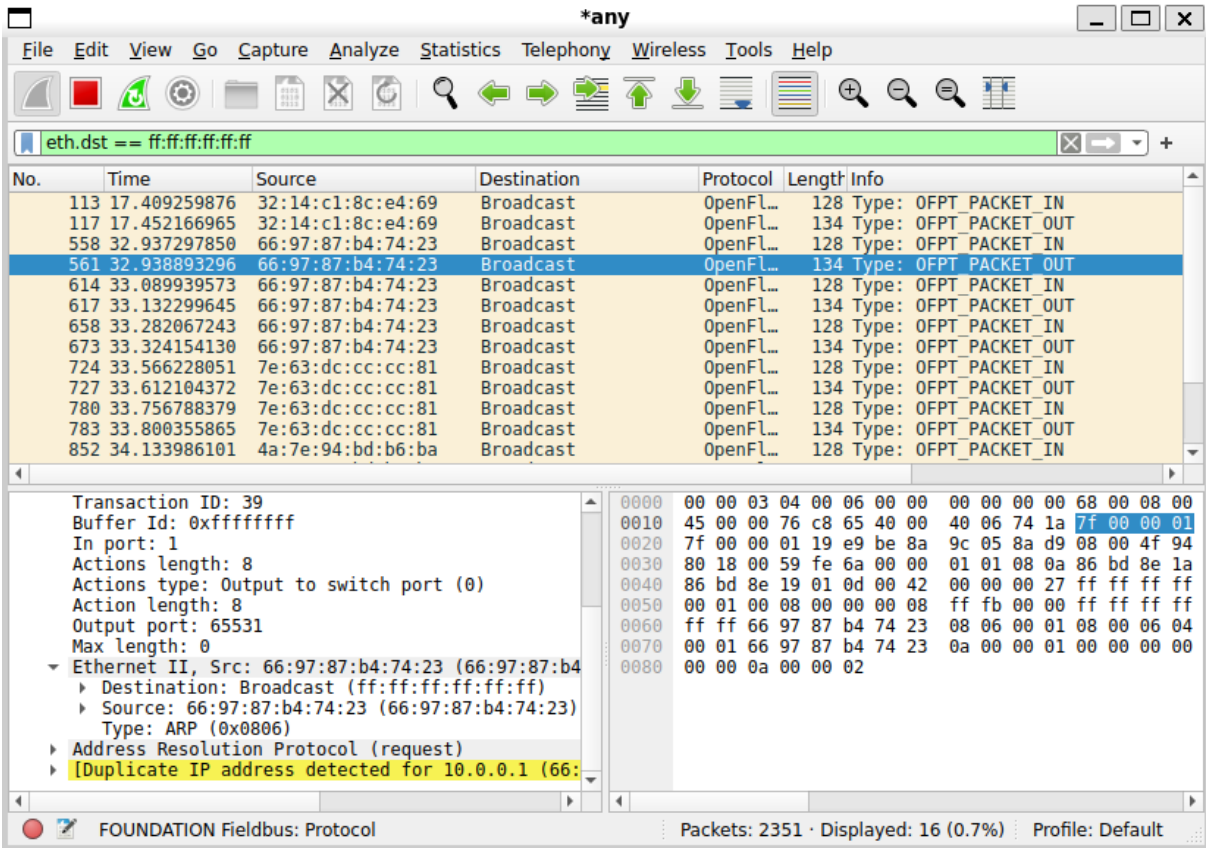
10.FLOW TABLE:

```
khushi@khushi: ~
khushi@khushi:~$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=23.388s, table=0, n_packets=0, n_bytes=0, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:ff:00:00:02 actions=FLOOD
 cookie=0x0, duration=23.261s, table=0, n_packets=0, n_bytes=0, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=33:33:ff:00:00:04 actions=FLOOD
 cookie=0x0, duration=23.230s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=23.133s, table=0, n_packets=0, n_bytes=0, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=33:33:ff:00:00:03 actions=FLOOD
 cookie=0x0, duration=22.909s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=22.782s, table=0, n_packets=1, n_bytes=90, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=22.781s, table=0, n_packets=2, n_bytes=140, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=22.366s, table=0, n_packets=2, n_bytes=140, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=22.238s, table=0, n_packets=2, n_bytes=140, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=22.110s, table=0, n_packets=1, n_bytes=90, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=22.109s, table=0, n_packets=2, n_bytes=140, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=6.947s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tp
a=10.0.0.2,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.944s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01,arp_spa=10.0.0.2,arp_tp
a=10.0.0.1,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.805s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tp
a=10.0.0.3,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.762s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:01,arp_spa=10.0.0.3,arp_tp
a=10.0.0.1,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.623s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tp
a=10.0.0.4,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.578s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:01,arp_spa=10.0.0.4,arp_tp
a=10.0.0.1,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.343s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.2,arp_tp
a=10.0.0.3,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.298s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:02,arp_spa=10.0.0.3,arp_tp
a=10.0.0.2,arp_op=2 actions=FLOOD
 cookie=0x0, duration=6.159s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.2,arp_tp
a=10.0.0.4,arp_op=1 actions=FLOOD
 cookie=0x0, duration=6.113s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:02,arp_spa=10.0.0.4,arp_tp
a=10.0.0.2,arp_op=2 actions=FLOOD
 cookie=0x0, duration=5.783s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.3,arp_tp
a=10.0.0.4,arp_op=1 actions=FLOOD
 cookie=0x0, duration=5.738s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:03,arp_spa=10.0.0.4,arp_tp
a=10.0.0.3,arp_op=2 actions=FLOOD
 cookie=0x0, duration=1.822s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01,arp_spa=10.0.0.2,arp_tp
a=10.0.0.1,arp_op=1 actions=FLOOD
 cookie=0x0, duration=1.820s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:02,arp_spa=10.0.0.1,arp_tp
a=10.0.0.2,arp_op=2 actions=FLOOD
 cookie=0x0, duration=1.566s, table=0, n_packets=0, n_bytes=0, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:01,arp_spa=10.0.0.3,arp_tp
```

11.WIRESHARK:


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.dst == ff:ff:ff:ff:ff:ff
No. Time Source Destination Protocol Length Info
2 0.000708025 00:00:00 00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
10 1.025162867 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
13 2.049330292 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
16 3.072826477 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
19 4.097265272 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
22 5.120814319 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
29 6.144876471 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
32 7.168794356 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
35 8.192790320 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
38 9.217236577 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
41 10.241327552 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
48 11.265212926 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
51 12.288905316 00:00:00_00:00:01 Broadcast OpenFl… 128 Type: OFPT_PACKET_IN
Frame 2: 128 bytes on wire (1024 bits), 128 bytes cap
Linux cooked capture v1
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127
Transmission Control Protocol, Src Port: 33136, Dst P
OpenFlow 1.0
0000 00 00 03 04 00 06 00 00 00 00 00 00 00 00 08 00
0010 45 c0 00 70 b3 cc 40 00 40 06 87 f9 7f 00 00 01
0020 7f 00 00 01 81 70 19 e9 1f fe 94 bd 9b c0 e1 52
0030 80 18 00 56 fe 64 00 00 01 01 08 0a 86 b3 6c 6f
0040 86 b3 68 70 01 0a 00 3c 00 00 00 00 ff ff ff ff
0050 00 2a 00 01 00 00 ff ff ff ff ff ff 00 00 00 00
0060 00 01 08 06 00 01 08 00 06 04 00 01 00 00 00 00
0070 00 01 0a 00 00 01 00 00 00 00 00 00 0a 00 00 ff
FOUNDATION Fieldbus: Protocol
Packets: 1457 · Displayed: 40 (2.7%)
Profile: Default


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
587 32.988457567 10.0.0.1 10.0.0.2 ICMP 100 Echo (ping) request id=0x0e19,
588 32.989314891 10.0.0.1 10.0.0.2 OpenFl… 184 Type: OFPT_PACKET_IN
599 33.032232882 10.0.0.1 10.0.0.2 OpenFl… 190 Type: OFPT_PACKET_OUT
601 33.032687691 10.0.0.1 10.0.0.2 ICMP 100 Echo (ping) request id=0x0e19,
602 33.032696909 10.0.0.1 10.0.0.2 ICMP 100 Echo (ping) request id=0x0e19,
603 33.032702439 10.0.0.1 10.0.0.2 ICMP 100 Echo (ping) request id=0x0e19,
604 33.032771334 10.0.0.2 10.0.0.1 ICMP 100 Echo (ping) reply id=0x0e19,
605 33.033477974 10.0.0.2 10.0.0.1 OpenFl… 184 Type: OFPT_PACKET_IN
608 33.080286867 10.0.0.2 10.0.0.1 OpenFl… 190 Type: OFPT_PACKET_OUT
610 33.080739150 10.0.0.2 10.0.0.1 ICMP 100 Echo (ping) reply id=0x0e19,
611 33.080755051 10.0.0.2 10.0.0.1 ICMP 100 Echo (ping) reply id=0x0e19,
612 33.080760852 10.0.0.2 10.0.0.1 ICMP 100 Echo (ping) reply id=0x0e19,
631 33.177169879 10.0.0.1 10.0.0.3 ICMP 100 Echo (ping) request id=0x0e2a,
Frame 587: 100 bytes on wire (800 bits), 100 bytes
Linux cooked capture v1
Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10
Internet Control Message Protocol
Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x2252 [correct]
[Checksum Status: Good]
Identifier (BE): 3609 (0x0e19)
Identifier (LE): 6414 (0x190e)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[No response seen]
Timestamp from icmp data: Apr 12, 2026 03:53:23.
0000 00 03 00 01 00 06 66 97 87 b4 74 23 00 00 08 00
0010 45 00 00 54 f4 07 40 00 40 01 32 9f 0a 00 00 01
0020 0a 00 00 02 08 00 22 52 0e 19 00 01 33 17 db 69
0030 00 00 00 00 f9 3f 01 00 00 00 00 00 10 11 12 13
0040 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23
0050 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33
0060 34 35 36 37
Internet Control Message Protocol: Protocol
Packets: 1374 · Displayed: 224 (16.3%)
Profile: Default



12.iperf result:

```
mininet> h2 iperf -s &
mininet> h1 iperf -c 10.0.0.2
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 35446 connected with 10.0.0.2 port 5001 (icwnd/mss/irtt=14/1448/51188)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0161 sec  5.60 GBytes  4.80 Gbits/sec
mininet>
```